CUSTOMER SUPPORT

Product Knowledge Base & Process Playbooks

RAG-ready reference document | 5 products | workflows, troubleshooting, and answer examples

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How to use this document

Upload this searchable PDF into a retrieval-augmented generation (RAG) or customer-support knowledge system. Each product has stable section IDs, terminology, workflow steps, troubleshooting guidance and example question-answer pairs. When answering, retrieve relevant sections, cite the PDF page, and distinguish guidance from verified product-specific configuration.

Important scope and safety

This is synthetic educational reference content, not vendor documentation or a verified runbook. Menu labels, features, permissions and exact behavior vary by product edition, deployment, version and customer configuration. Validate any operational procedure against the organization's approved documentation before production use. Never ask users to disclose passwords, tokens, private customer data or sensitive logs.

Product index

WCP - Windchill PLM: Product lifecycle management platform for controlled product structures, engineering documents, revisions, changes and cross-functional collaboration.

CRE - Creo: Computer-aided design software used to model parts, assemble components, create drawings and manage engineering design intent.

THW - ThingWorx: Industrial IoT application platform for connecting assets, collecting telemetry, modeling entities and presenting operational dashboards.

CDB - Codebeamer: Application lifecycle management platform for requirements, risks, tests, defects and development traceability.

VUF - Vuforia: Augmented reality and connected-workforce technology used to deliver contextual instructions and visual guidance in industrial scenarios.

Suggested retrieval metadata

document_type=product_knowledge_base; version=1.0; content_status=illustrative; product_id=[WCP|CRE|THW|CDB|VUF]; section_type=[overview|concepts|workflow|troubleshooting|faq]; source=customer_support_product_knowledge_base.pdf

Global support intake and response process [GEN-01]

1. Identify product, version/build, deployment model and business impact.
2. Record the user goal, expected result, actual result, timestamp and reproducible steps.
3. Ask for sanitized error text, relevant log excerpts and screenshots without secrets or personal data.
4. Check service status, access scope and whether the issue is limited to one user or environment.
5. Retrieve product-specific knowledge; prefer exact product/version matches and approved internal runbooks.
6. Propose the lowest-risk diagnostic action first; seek authorization before changes or restarts.
7. Confirm resolution with the requester and document root cause, workaround and evidence.

Severity classification [GEN-02]

P1 Critical: Production outage, safety impact or major business disruption; escalate immediately under incident policy.

P2 High: Major feature unavailable with significant impact and no viable workaround.

P3 Medium: Partial functionality or limited impact with a workable alternative.

P4 Low: How-to request, cosmetic defect or minor inconvenience.

Answer quality rubric [GEN-03]

A good answer identifies the correct product, addresses the question directly, offers numbered steps when appropriate, includes prerequisites and validation checks, names uncertainty, and cites the supporting page. Do not invent exact menus, commands, error codes, release dates, API endpoints or configuration paths.

Windchill PLM | Product overview [WCP-00]

Product lifecycle management platform for controlled product structures, engineering documents, revisions, changes and cross-functional collaboration.

Core concepts [WCP-01]

Part: A managed representation of a physical or logical product component.

CAD document: Managed design data associated with engineering tools.

BOM: Bill of materials describing component relationships and quantities.

Lifecycle state: A controlled status such as In Work, Under Review or Released; actual states vary.

Change notice: A controlled change record coordinating approved implementation tasks.

Typical support scope

Access and permissions; workflow or data issues; version and configuration questions; integration behavior; performance symptoms; troubleshooting and knowledge retrieval. Record exact deployment details before recommending changes.

Workflow 1: Create and review a part [WCP-P01]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Confirm the user has access to the correct product or library context.
2. Open the relevant context and navigate to the part-management area.
3. Create a part with a unique number or configured auto-numbering, name, unit and classification.
4. Complete required attributes and associate documents if applicable.
5. Save and validate the part number, revision, ownership and lifecycle state.
6. Route for review or release using the organization's configured workflow.

Expected outcome: Part record exists, required metadata is populated, and lifecycle reflects approved status.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 2: Manage a product BOM [WCP-P02]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Open the parent assembly part at the correct revision and view.
2. Review the existing structure and confirm intended effectivity/configuration.
3. Add or replace child parts and set quantities and units.
4. Check duplicates, missing components, reference designators and circular structures.
5. Save and compare the new structure against the prior baseline.
6. Submit the revised structure through the applicable change-control process.

Expected outcome: BOM is traceable to a revision and approved change record.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 3: Process an engineering change [WCP-P03]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Log the issue and affected parts/documents with business impact.
2. Create or associate a change request, where configured.
3. Evaluate scope, risk, approvers and proposed solution.
4. Create a change notice and assign implementation tasks.
5. Update impacted objects, perform review and validate the change.
6. Approve, release and communicate the new baseline.

Expected outcome: Approved change history links affected objects, tasks and release decisions.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Troubleshooting guide [WCP-T01]

1. Symptom: Cannot check in CAD data. Recommended checks: Check workspace state, ownership, checkout status, file conflicts and server connectivity; preserve the local workspace before retrying.

2. Symptom: User cannot see a part. Recommended checks: Verify context, access policy, team role, lifecycle permissions and filters; do not bypass permissions.

3. Symptom: Wrong BOM revision appears. Recommended checks: Check view, revision, configuration specification, effectivity and cached search filters.

4. Symptom: Workflow is stuck. Recommended checks: Identify current task owner, routing conditions, required approvals and workflow errors; escalate to the workflow administrator.

Example customer questions and grounded answers [WCP-FAQ]

Q1: How do I create a new part?

A: Open the correct product context, create a part, enter mandatory attributes, save, then follow the configured review workflow.

Q2: Why is my BOM missing components?

A: Check the configuration specification, revision, view, effectivity and user permissions before assuming data is absent.

Q3: What is the difference between revision and iteration?

A: A revision typically marks a formal design level; an iteration tracks updates within a revision. Exact rules depend on configuration.

Q4: Can I release a part without approval?

A: Follow the configured lifecycle and authorization policy. Do not bypass required approvals.

Escalation notes [WCP-ESC]

Escalate when the user cannot proceed after safe checks, when an administrative change is required, or when data integrity, security, compliance or production availability may be affected. Include a sanitized reproduction, product version, affected object IDs where permitted, impact, time window and diagnostics.

Creo | Product overview [CRE-00]

Computer-aided design software used to model parts, assemble components, create drawings and manage engineering design intent.

Core concepts [CRE-01]

Parametric feature: Geometry controlled by dimensions, constraints and feature history.

Assembly constraint: Relationship that positions components relative to references.

Regeneration: Recalculation of model geometry after changes.

Family table: Mechanism for managing configured variants in supported workflows.

Drawing: 2D manufacturing or communication representation linked to 3D models.

Typical support scope

Access and permissions; workflow or data issues; version and configuration questions; integration behavior; performance symptoms; troubleshooting and knowledge retrieval. Record exact deployment details before recommending changes.

Workflow 1: Build a parametric part [CRE-P01]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Create a new part using the correct template and unit system.
2. Sketch the base profile and apply dimensions and constraints.
3. Create solid features such as extrude, revolve, holes and rounds.
4. Rename critical features and add material and parameters as required.
5. Regenerate and inspect model failures or unresolved references.
6. Save with an approved filename and validate the mass properties.

Expected outcome: A regenerating part model with documented dimensions and design intent.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 2: Create and validate an assembly [CRE-P02]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Start an assembly using the correct template and coordinate convention.
2. Insert the base component and establish placement.
3. Add components and apply appropriate placement constraints.
4. Regenerate and check degrees of freedom and interference.
5. Verify assembly structure and component versions.
6. Save and prepare the design for downstream drawing or PLM check-in.

Expected outcome: Stable assembly with resolved placement and validated structure.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 3: Prepare a manufacturing drawing [CRE-P03]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Open the correct approved model and drawing template.
2. Place orthographic, section or detail views as needed.
3. Add critical dimensions, tolerances, notes and material information.
4. Verify view scale, annotations and drawing standards.
5. Regenerate and check for missing references or outdated views.
6. Export the approved output and store it in the controlled repository.

Expected outcome: Readable drawing aligned with the model and review process.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Troubleshooting guide [CRE-T01]

1. Symptom: Regeneration failed. Recommended checks: Inspect failed features and missing references; repair upstream dependencies before downstream features.

2. Symptom: Assembly component is underconstrained. Recommended checks: Review placement constraints and datum references; avoid unnecessary conflicting constraints.

3. Symptom: Drawing views are outdated. Recommended checks: Regenerate the source model and drawing, verify references and check that the correct model revision is loaded.

4. Symptom: Unexpected model scale. Recommended checks: Check units, imported geometry settings and template configuration before editing dimensions.

Example customer questions and grounded answers [CRE-FAQ]

Q1: What causes regeneration errors?

A: Common causes include broken references, invalid geometry, conflicting constraints and upstream feature changes.

Q2: How do I create an assembly?

A: Create an assembly, place a base component, add components with constraints, regenerate and validate interference.

Q3: How can I avoid broken references?

A: Prefer stable datums and intentional design references, and review dependencies when changing upstream features.

Q4: Can Creo drawings update from models?

A: Associative drawings can reflect model changes after regeneration; validate the drawing before release.

Escalation notes [CRE-ESC]

Escalate when the user cannot proceed after safe checks, when an administrative change is required, or when data integrity, security, compliance or production availability may be affected. Include a sanitized reproduction, product version, affected object IDs where permitted, impact, time window and diagnostics.

ThingWorx | Product overview [THW-00]

Industrial IoT application platform for connecting assets, collecting telemetry, modeling entities and presenting operational dashboards.

Core concepts [THW-01]

Thing: A modeled asset or entity with properties, services and events.

Thing Template: Reusable definition for a family of similar assets.

Property: A value such as temperature, status or operating hours.

Service: An executable operation exposed by an entity.

Mashup: A UI composition used to display and interact with data.

Typical support scope

Access and permissions; workflow or data issues; version and configuration questions; integration behavior; performance symptoms; troubleshooting and knowledge retrieval. Record exact deployment details before recommending changes.

Workflow 1: Onboard a connected asset [THW-P01]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Define the business asset, identifiers and telemetry requirements.
2. Create or select a Thing Template and required properties.
3. Create the Thing instance and configure the authorized connection method.
4. Map incoming telemetry fields to the expected properties and data types.
5. Test with non-production data and verify timestamps, units and update rates.
6. Apply access controls and document the asset owner and monitoring plan.

Expected outcome: Asset properties update correctly and are visible to authorized users.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 2: Build an asset monitoring dashboard [THW-P02]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. List the KPIs, alarm thresholds and user roles.
2. Create or reuse entities, data services and property bindings.
3. Design a Mashup with status indicators, trend charts and alert lists.
4. Bind each widget to validated data and handle missing or stale values.
5. Test role permissions, refresh behavior and performance.
6. Publish to the intended audience and document operational ownership.

Expected outcome: Users can see current and historical asset conditions with clear freshness indicators.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 3: Investigate an IoT alert [THW-P03]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Record asset, alert time, severity and observed values.
2. Check whether the telemetry is current and correctly time-stamped.
3. Compare values against configured thresholds and recent trends.
4. Inspect connectivity and data quality before treating the alert as a physical fault.
5. Escalate operational hazards through the site's safety procedure.
6. Document findings and close only after validation.

Expected outcome: Alert investigation includes evidence, owner and resolution.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Troubleshooting guide [THW-T01]

1. Symptom: Telemetry stopped updating. Recommended checks: Check device connectivity, ingestion path, authentication, property bindings and timestamps.

2. Symptom: Mashup shows blank values. Recommended checks: Verify data service execution, property type, bindings, permissions and empty-result handling.

3. Symptom: Alerts trigger too frequently. Recommended checks: Review thresholds, hysteresis, signal noise and expected operating range with the process owner.

4. Symptom: Dashboard is slow. Recommended checks: Inspect query volume, refresh frequency, payload size and data retention/indexing choices.

Example customer questions and grounded answers [THW-FAQ]

Q1: What is a Thing Template?

A: A reusable model defining shared properties, services and behaviors for similar Things.

Q2: Why is my dashboard blank?

A: Check whether the data service returns values, whether bindings are correct and whether the user has access.

Q3: How can I monitor a machine?

A: Model the machine, connect telemetry, validate properties and build a Mashup with KPIs and alerts.

Q4: How should missing sensor values appear?

A: Display a missing or stale-data state rather than presenting the last known value as current.

Escalation notes [THW-ESC]

Escalate when the user cannot proceed after safe checks, when an administrative change is required, or when data integrity, security, compliance or production availability may be affected. Include a sanitized reproduction, product version, affected object IDs where permitted, impact, time window and diagnostics.

Codebeamer | Product overview [CDB-00]

Application lifecycle management platform for requirements, risks, tests, defects and development traceability.

Core concepts [CDB-01]

Tracker: A configurable collection of work items such as requirements or defects.

Work item: An individual managed record with fields, status and relationships.

Baseline: A captured reference point for a defined set of artifacts.

Traceability: Links between requirements, tests, risks and implementation records.

Test run: Execution record containing outcomes and evidence.

Typical support scope

Access and permissions; workflow or data issues; version and configuration questions; integration behavior; performance symptoms; troubleshooting and knowledge retrieval. Record exact deployment details before recommending changes.

Workflow 1: Capture and approve a requirement [CDB-P01]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Select the correct project and requirements tracker.
2. Create a uniquely identifiable requirement with a clear statement and rationale.
3. Record priority, owner, acceptance criteria and relevant safety or regulatory classification.
4. Link the requirement to upstream stakeholder needs and downstream tests.
5. Run review, resolve comments and record approval evidence.
6. Baseline the approved set using the organization's procedure.

Expected outcome: Requirement has clear acceptance criteria, ownership and traceable approval.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 2: Execute a test cycle [CDB-P02]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Identify the approved test set and intended software/hardware baseline.
2. Confirm environment, test data and prerequisites.
3. Create or open the test run and assign testers.
4. Execute steps and record pass/fail outcomes with evidence.
5. Log defects and link them to failed steps and requirements.
6. Review completion, coverage and unresolved risks before sign-off.

Expected outcome: Auditable test results and linked defects support release decisions.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 3: Manage a defect to closure [CDB-P03]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Record observed behavior, expected behavior, environment and reproduction steps.
2. Classify severity, priority and affected release.
3. Link the defect to related requirements, tests and change records.
4. Assign an owner and implement a reviewed fix.
5. Retest the fix and perform relevant regression tests.
6. Close with evidence or reopen if acceptance criteria remain unmet.

Expected outcome: Defect status reflects verified outcome and traceable evidence.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Troubleshooting guide [CDB-T01]

1. Symptom: Missing requirement-to-test links. Recommended checks: Check tracker link types, filters, item permissions and baseline scope.

2. Symptom: Test case cannot be executed. Recommended checks: Confirm test configuration, assignment, prerequisite state and tester permissions.

3. Symptom: Unexpected workflow transition block. Recommended checks: Inspect mandatory fields, status permissions, approval gates and configured guards.

4. Symptom: Baseline seems incomplete. Recommended checks: Verify included trackers, filters, snapshot scope and effective baseline version.

Example customer questions and grounded answers [CDB-FAQ]

Q1: How do I trace requirements to tests?

A: Create approved relationships between requirements and test cases, then inspect the traceability view or report.

Q2: What is a baseline used for?

A: It captures a reference state for review, audit, comparison and release evidence.

Q3: How do I report a defect?

A: Provide reproduction steps, expected versus actual results, environment, severity and supporting evidence.

Q4: Can a failed test be closed without a fix?

A: Follow the documented disposition and approval process; retain the rationale and evidence.

Escalation notes [CDB-ESC]

Escalate when the user cannot proceed after safe checks, when an administrative change is required, or when data integrity, security, compliance or production availability may be affected. Include a sanitized reproduction, product version, affected object IDs where permitted, impact, time window and diagnostics.

Vuforia | Product overview [VUF-00]

Augmented reality and connected-workforce technology used to deliver contextual instructions and visual guidance in industrial scenarios.

Core concepts [VUF-01]

Experience: An authored AR or assisted-workflow presentation for a user.

Target: A reference used to associate an experience with an object or environment, depending on product and configuration.

Step: A discrete instruction or activity in a guided workflow.

Spatial alignment: Placement of digital guidance relative to physical equipment.

Published content: A reviewed experience available to intended users.

Typical support scope

Access and permissions; workflow or data issues; version and configuration questions; integration behavior; performance symptoms; troubleshooting and knowledge retrieval. Record exact deployment details before recommending changes.

Workflow 1: Design an equipment guidance experience [VUF-P01]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Define the task, intended audience, safety constraints and device capabilities.
2. Gather approved procedures, images, CAD or 3D assets as appropriate.
3. Break the procedure into short steps with unambiguous success criteria.
4. Author visual cues, labels and step transitions in the supported authoring tool.
5. Validate alignment, readability and behavior on target hardware.
6. Review with the equipment owner and publish through controlled access.

Expected outcome: Approved experience is understandable, aligned and safe for its intended environment.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 2: Validate an AR experience in the field [VUF-P02]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Confirm the approved content version and compatible device.
2. Check lighting, visibility, network conditions and target recognition.
3. Launch the experience in the designated test area.
4. Walk through each step and verify instructions against the approved procedure.
5. Record alignment issues, incorrect text and user feedback.
6. Resolve issues and repeat acceptance testing before rollout.

Expected outcome: Field test evidence demonstrates correct guidance and user comprehension.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Workflow 3: Maintain a published instruction [VUF-P03]

Prerequisites: authorized access, appropriate environment, approved process and correct object or asset context.

1. Receive a change request and identify affected equipment or steps.
2. Review the updated approved work instruction.
3. Revise media, labels and interactions while retaining version history.
4. Perform technical and safety reviews and device regression tests.
5. Publish the approved version and notify affected users.
6. Archive or retire obsolete guidance according to policy.

Expected outcome: Users access the current approved procedure with traceable change history.

Validation: verify record state, expected output, access rights and traceable evidence before closing the request.

Troubleshooting guide [VUF-T01]

1. Symptom: Target is not recognized. Recommended checks: Check target quality, device camera, lighting, distance and supported tracking conditions.

2. Symptom: AR overlay is misaligned. Recommended checks: Recalibrate placement, verify target/anchor configuration and retest on the physical equipment.

3. Symptom: Experience fails to load. Recommended checks: Check user access, content publication, network connectivity and device compatibility.

4. Symptom: Instructions differ from procedure. Recommended checks: Stop use for the affected task, compare against approved documentation and request a controlled update.

Example customer questions and grounded answers [VUF-FAQ]

Q1: What is Vuforia used for?

A: It can deliver contextual AR instructions, visualization and guided industrial work experiences.

Q2: Why does an overlay drift?

A: Tracking quality, lighting, anchor placement or environmental conditions may affect alignment.

Q3: How do I publish an instruction?

A: Author and test the experience, obtain required approvals, then publish to authorized users.

Q4: Can AR replace a safety procedure?

A: No. Follow site safety requirements and approved procedures; AR is supporting guidance.

Escalation notes [VUF-ESC]

Escalate when the user cannot proceed after safe checks, when an administrative change is required, or when data integrity, security, compliance or production availability may be affected. Include a sanitized reproduction, product version, affected object IDs where permitted, impact, time window and diagnostics.

Cross-product workflows [XPR-01]

Creo to Windchill: controlled CAD handoff

1. Create and validate the CAD model in Creo.
2. Confirm the correct workspace, target context and integration setup.
3. Associate or create the corresponding managed CAD document in Windchill.
4. Check in the authorized model and inspect references and related part associations.
5. Validate BOM or drawing dependencies and submit for change or release review.

Codebeamer to engineering verification

1. Define a testable requirement and acceptance criteria in Codebeamer.
2. Link the requirement to design evidence or implementation records using approved integration mechanisms.
3. Create test cases, execute tests and attach results.
4. Record failures as defects and verify corrective actions.
5. Review traceability and approved baselines before release.

ThingWorx and Vuforia: contextual operations

1. Model the asset and telemetry in ThingWorx and validate data quality.
2. Identify the approved task or operational state that needs guidance.
3. Author a Vuforia experience using approved procedures.
4. Connect contextual data only through supported, secured integration.
5. Validate the experience in the real environment and monitor for stale or incorrect data.

Knowledge base ingestion guidance [RAG-01]

Recommended chunking: split by product heading and workflow or FAQ subsection, with approximately 350-700 words per chunk and 50-100 words overlap. Preserve page number, section ID, product name, version and document title in metadata. Do not merge chunks from different products if avoidable.

Retrieval: filter by detected product first when confident, retrieve 3-6 relevant chunks, and require evidence for each substantive answer. If the query lacks a product or version, ask a clarifying question. For unsupported details, state that the document does not specify them and route to approved product documentation.

Suggested test queries [RAG-02]

- How do I create a part and release it in Windchill?
- Why is my Windchill BOM showing the wrong revision?
- What steps are needed to make a Creo assembly?
- Why did my Creo model fail regeneration?
- How do I onboard a machine into ThingWorx?
- What causes blank values in a ThingWorx Mashup?
- How can I trace a Codebeamer requirement to a test?
- How do I close a defect with verification evidence?
- How do I validate a Vuforia AR instruction on site?

- What should I check if an AR overlay is misaligned?
- What information is required to escalate a P1 incident?
- How can Creo CAD models be handed off to Windchill?

Sample answer policy [RAG-03]

Answer format: Product; direct answer; numbered steps; prerequisites/risks; validation; source document and page number. Do not claim an accuracy percentage or knowledge coverage metric based solely on this reference PDF. Those metrics require a separate evaluation dataset and log instrumentation.

Document maintenance [RAG-04]

For production deployment, replace illustrative procedures with approved version-specific content, assign a document owner, record review and expiry dates, include links to authoritative internal manuals, and retest answers after each revision. Exclude credentials, confidential data and raw unsanitized production logs.